Isaac Palma-Medina

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Experience

Student Researcher & Teaching Assistant Mar. 2026 — Present
University of Luxembourg (UniLu)

- **GPU Optimisation (main research focus):** Investigating GPU parallelisation of Cost Function Network (CFN) solving.
- **HPC Performance Characterisation:** Developing a ReFrame-based framework to characterise hardware-specific cluster partitions through queue-based measurements, collaborating with teams in Luxembourg and Costa Rica.
- **AI for Environmental Monitoring:** Applying unsupervised learning to anomaly detection and data-quality analysis across a weather-station network in Costa Rica, in collaboration with the University of Costa Rica.
  - "Cross-Method Agreement in Unsupervised Anomaly Detection across a Distributed Meteorological Sensor Network in Costa Rica" — IEEE BioInspired Processing 2026.
  - "Public Emergency Alerts as Weak Labels for Weather Station Anomaly Detection: Evaluating an LSTM Autoencoder in Costa Rica" — Preprint.
- **Teaching Assistant — Parallel Computing:** Reviewing student solutions and providing in-class technical support.

Collaborator Jul. 2024 — Sept. 2025
LHCb Experiment, European Organization for Nuclear Research (CERN)

- Co-developed the reconstruction pipeline for the full Ring Imaging Cherenkov (RICH) subdetectors, migrating processing from CPU-based code to a GPU-optimized CUDA environment within LHCb's real-time trigger system.
- Conducted track-resolution studies between trigger stages to evaluate GPU-based reconstruction performance and support its integration and optimization.
- **Summer Student Program** (Jul. 2024 — Aug. 2024):
  - Developed monitoring algorithms to assess particle-track reconstruction resolution and data quality within the trigger system.
  - Performed scientific data analysis and visualization in Python during a two-month research stay at CERN.

Research Assistant Sept. 2024 — Sept. 2025
National Collaboratory for Advanced Computing (CNCA), CeNAT-CONARE

- Conducted research supporting the CERN LHCb collaboration through Costa Rica's institutional participation in the experiment.

Software Developer Jun. 2024 — Nov. 2024
University of Costa Rica (UCR)

- Developed a digital emergency-call management system that replaced previously manual workflows.

Software Developer & Scrum Master Mar. 2023 — Jun. 2024
9-1-1 Emergency System of Costa Rica

- Led development of a real-time staff scheduling system valued at over $50,000 for a national emergency-services environment.

Education

University of Luxembourg (UniLu) Sept. 2025 — Present
Master's Degree in High Performance Computing

- UniLu International Awards 2026 awardee (Ministry of Foreign Affairs).
- Full scholarship awarded by the Government of Luxembourg for academic merit and international collaboration in HPC.
- Participant in the European Master for High Performance Computing (EUMaster4HPC) track.

National University of Costa Rica (UNA) 2021 — Jun. 2025
Bachelor's Degree in Information Systems Engineering (incl. Associate Degree in Software Development, 2023)

- 2024 Distinguished Student Award — highest GPA in the program; academic excellence scholarship (9.61/10).
- Student tutor in C++ and object-oriented programming for two years; completed parallel computing electives.

Johns Hopkins University (JHU) 2024 — 2025
Non-Credit Specialization in GPU Programming (CUDA), authorized by JHU

Selected Talks & Activities

Invited Speaker — Inaugural Talk Series | IEEE Student Branch, UNA May 2026
- Delivered "From International Experience to Research: HPC and AI from Costa Rica," introducing HPC and ongoing research projects to computer science students.

Speaker — "International Opportunities for Costa Rican Students" | UNA May 2026
- Presented international academic and research opportunities, highlighting Luxembourg–Costa Rica collaboration and HPC opportunities at UniLu.

Instructor — Introduction to High Performance Computing Workshop | UNA Feb. 2025, Apr. 2026
- Introduced HPC concepts and guided over 60 participants through hands-on supercomputer exercises across two editions.

Participant — Costa Rica Big Data School | CNCA Sept. 2025
- Completed hands-on training in Python and AI-driven data analysis using FRONTERA (TACC), covering the computational research workflow.

Teaching Assistant — Parallel Programming | LEAD University Jun. 2025 – Sept. 2025
- Evaluated assignments and provided technical feedback to students.

Participant & Local Organizing Support — CLASHEP 2025 | CERN May 2025
- Attended particle-physics lectures at the CERN Latin-American School of High-Energy Physics and supported the local organizing committee.

Participant (1st Place, Team Challenge) — Costa Rica HPC School | CNCA Jan. 2025
- Trained in Kokkos and MPI; competed in a team event of three computational optimization challenges.

Participant — Physics Without Frontiers Workshop | ICTP Dec. 2024
- Analyzed B-meson decays and CP violation using open LHCb data with Python.

Workshop Assistant — HPC and Kokkos Workshop, 42nd IEEE CONCAPAN | IEEE Nov. 2024
- Supported the workshop's delivery and participated in conference sessions.

Instructor — Free Community Python Course | Cisco NetAcad & UNA Nov. 2023 – Feb. 2024
- Organized and delivered a Python programming course for over 100 community participants.

Languages

- Spanish — Native
- English — C1 (TOEFL Essentials)
- French — Beginner